Miss H × VEVFM — Reading the earliest instability in sepsis: Registration 2a (eICU, single held-out run)

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Summary

This registration fixes, before any eICU outcome is seen, one ward alarm and one primary claim: at the same false-alarm cost on recovered patients, the frozen alarm warns a larger share of septic patients whose first escalation is a vasopressor (the vasopressor-first group) than full NEWS does.

- **Test data:** eICU Collaborative Research Database, septic first ICU stays, run **once** after signing.
- **Calibration data:** MIMIC-III (all tuning, rehearsal and line-setting happened here; nothing there counts as evidence).
- **Alarm:** amber from two readings (a pressure "leading channel" and S/F, with a breathing-effort fallback when oxygen is not recorded); red when amber is still present at a 3-hour re-check.
- **Primary test:** one-sided exact McNemar on the vasopressor-first group, frozen alarm vs full NEWS at matched cost, one-sided $p < 0.025$.
- **Everything else** (ventilation-first, combined and early-crash groups, whole-ward total, red, S/F-forward versions) is a pre-specified secondary or a descriptive report, published whatever it shows.

Aim: the earliest instability in sepsis

The aim is to see a septic patient's slide toward shock or respiratory failure while he still looks stable, so the team has hours, not minutes, to act.

The problem. In sepsis, a patient can move from a stable-looking ICU bed to needing a vasopressor or a ventilator within hours. Compensation hides the slide: heart rate, vessel tone and supplemental oxygen hold the familiar numbers near normal while reserve is being used up. Early-warning scores such as NEWS add points only once values leave normal bands, so a patient can be close to the edge with a modest score, and a patient held at SpO2 95 on a mask can look like one breathing room air.

What the architecture reads. For every patient, every hour: where he sits between the state of recovered patients and the state at which vasopressors usually start; which channel is carrying him there (pressure, or breathing read against the oxygen it took); and, after a warning and treatment, whether he comes back. The earliest instability is the point where one channel has held near that edge for three hours.

What this registration tests. Whether that reading warns more septic patients who later need a vasopressor than NEWS does, at the same false-alarm cost, in a held-out cohort from 204 hospitals. It does not claim an earlier warning on the clock, and it does not claim that warning prevents escalation; that needs a prospective study.

Sepsis definition. The cohort is defined by an ICU admission diagnosis of sepsis (eICU APACHE diagnosis), not by Sepsis-3 criteria computed from the record; this is stated as a limit.

Scope. This registration tests the framework in sepsis only and claims nothing beyond it. The framework itself (position between a recovered state and an escalation state, a leading channel, breathing read against the oxygen it took, and a re-check after action) does not depend on the cause of deterioration; applying it to other populations (for example post-operative bleeding, heart failure, pneumonia, or general wards) requires anchors and lines set on that population and its own pre-registered test.

Background

Registration 1 (signed 3 Nov 2026, MIMIC-IV) was null over NEWS: reading each patient's movement from his own starting point did not help, because about 94% of deteriorating patients stayed within their own range until about the final hour.

The design was rebuilt on MIMIC-III with five changes:

Registration 1	Registration 2a
Distance from each patient's own start	Position between two fixed ends: where recovered patients sit, and where patients are when a vasopressor starts
All channels averaged	The most strained channel speaks (leading channel), if it holds for 3 hours
SpO2 read on its own	S/F: SpO2 read against the oxygen it took
One score for every kind of failure	Two readings at once (pressure, breathing) under one false-alarm budget
No step after a warning	Re-check 3 hours after amber; still amber = red

Data, groups and definitions

The test cohort is every adult septic first ICU stay in eICU (APACHE admission diagnosis containing "sepsis"): 19,514 stays from 204 hospitals, each placed in exactly one group by its first escalation (counts only; no outcomes viewed).

Group	Definition	Stays	Code label
Recovered without escalation	No vasopressor or ventilation; discharged alive	9,905	Ajay
Vasopressor-first deterioration	Vasopressor first, escalation at 5 h or later	993	Kranthi
Ventilation-first deterioration	Ventilation first, escalation at 5 h or later	876	Sundar
Combined deterioration	Vasopressor and ventilation within 1 h of each other, at 5 h or later	42	mixed
Early crash	Escalation within 3 h of ICU arrival, not after surgery or recovery	2,489	Naveen
Excluded: arrived on support	Escalation at or before ICU arrival	3,578	-
Excluded: died without escalation	No escalation, died in hospital	976	-
Excluded: discharge status missing	No escalation, survival unknown	140	-
Excluded: grey zone	Escalation between 3 and 5 h	507	-
Excluded: planned early escalation	Escalation within 3 h after surgery or recovery	8	-

Escalation time (tip) = earliest of: first vasopressor infusion; ventilator start in the respiratory-care table; mechanical ventilation in the treatment table; a tube or ventilator charted as the oxygen device. Ventilator settings alone are not used (they also appear with non-invasive support).

Recovered patients' watch window: each recovered patient is watched to a pseudo-escalation time drawn at random (seed 2026) from the escalation times of the vasopressor-first, ventilation-first and combined groups (5-72 h), capped at his ICU stay. All windows end at 72 h.

Hourly values: eICU vitals (about every 5 minutes) are averaged into hourly values; cuff and arterial pressures are pooled. The last FiO2, flow and oxygen device in each hour, and the lowest GCS, are carried forward up to 12 h.

Oxygen mapping (FiO2 when not charted): room air 0.21; nasal cannula 0.21 + 0.04 × L/min (2 L if missing), max 0.44; masks and trach 0.40; non-rebreather 0.85; high-flow 0.60; BiPAP/CPAP 0.50; other text no estimate. A charted FiO2 overrides; "O2 L/%" values 1-15 are litres, 21-100 percent, 0-1 a fraction.

The frozen alarm

Amber fires in any hour where one of three readings crosses its line; the lines were set on MIMIC-III thinned to eICU's oxygen charting (56% of hours without oxygen information), at about 1 false amber per recovered patient per day.

Reading	Amber when	Used in
Pressure leading channel	Highest of the MAP, shock-index and pulse-pressure road positions, 3-h mean, >= 1.350	Every hour
S/F (SpO2 / FiO2)	<= 98.9	Hours with oxygen recorded
Breathing effort position	Mean of (RR - 19.0) / 10.0 and (97.0 - SpO2) / 4.7, 3-h mean, >= 1.060	Hours without oxygen recorded

Pressure road anchors: recovered end MAP 75, shock index 0.71, pulse pressure 58; vasopressor end MAP 58, shock index 0.96, pulse pressure 43. Each channel's position = (value - recovered end) / (vasopressor end - recovered end), clipped to -0.5..1.5. In bedside terms, amber needs one of: shock index about 1.05 or more, MAP about 52 or less, or pulse pressure about 38 or less, held for 3 hours.

Red (re-check): for every amber episode starting at hour t, red fires at t + 3 if amber is still present.

Comparator, full NEWS: NEWS points for breathing rate, SpO2, systolic pressure, heart rate and temperature (3 or more present), + 2 if FiO2 > 0.21, + 3 if the last GCS within 12 h is below 15; 3-h mean. Its line is set on eICU's recovered patients to the nearest value whose false-amber cost is at or below the frozen alarm's. The same rule sets NEWS's red line.

Primary claim and test

Claim (one-sided): at matched false-amber cost on recovered patients, the frozen alarm warns a larger share of eICU vasopressor-first patients before their escalation than full NEWS.

1. For each vasopressor-first patient: was amber present in any hour before escalation (frozen alarm: yes/no; full NEWS at its matched line: yes/no).
2. Count discordant patients: caught only by the frozen alarm (b) and only by NEWS (c).
3. One-sided exact McNemar test (binomial on b of b + c, p = 0.5).
4. **Supported** if the frozen alarm's share is higher and one-sided p < 0.025 (equivalent to the conventional two-sided 0.05; chosen to offset the selection of the best of 20 versions on MIMIC-III). Also reported: the paired difference in shares with a 95% confidence interval, and median hours from first amber to tip for each.

If not supported, the result is reported as not supported. No lines, groups, windows or definitions change after the run, and no further analysis is presented as planned.

Secondaries

Two key secondaries carry a stated direction; the rest are reported without a predicted winner and without multiplicity claims.

Secondary	Comparison	Direction
Re-check red, vasopressor-first group	Frozen alarm's red vs NEWS alone, NEWS line set to the same red cost	One-sided: frozen alarm higher
Re-check red vs NEWS re-check	Same 3-hour re-check given to NEWS; NEWS amber line set so its red cost is at or below the frozen alarm's	Two-sided
Whole-ward total	Vasopressor-first + ventilation-first + combined groups warned before escalation, frozen alarm vs full NEWS at matched cost	Two-sided
S/F-forward A	Pressure leading channel >= 1.5 or S/F <= 247.5 (from MIMIC-III check C16c), vs NEWS at that version's own cost	Two-sided; ventilation-first group reported by oxygen-recorded band

Secondary	Comparison	Direction
S/F-forward B	Pressure >= 1.5, or S/F <= 209 (oxygen recorded), or breathing rate 3-h >= 23.3 (oxygen not recorded) (from C16i), vs NEWS at its own cost	Two-sided; ventilation-first group by oxygen band
Effort-only check	Breathing read by effort alone, no S/F	Two-sided; shows whether results depend on oxygen charting
Alarms on improving patients	Share of amber episodes starting while the patient was improving (pressure falling, S/F rising, or breathing rate falling vs 3 h earlier)	Two-sided

All comparisons use the same paired approach as the primary where patients are shared. Two further descriptive reports, added before signing: **paired timing** (for patients warned by both the architecture and full NEWS: share where the architecture spoke first, in the same hour, or after NEWS, and the median hour difference, by group and for the ventilation-first group by hospital oxygen-charting band; for the frozen alarm and S/F-forward B) and **precision** (share of amber episodes that started on a patient who later escalated, before escalation).

Descriptive reports by group

Each report below prints from the same frozen script and is published as it comes out; none carries a claim.

Group	Reports
Vasopressor-first	Lead time; patients whose shock index stayed low (mean < 0.8 in the last 6 h); late crossers (averaged pressure road 3-h never >= 0.53, or < 2 h before escalation); first reading to speak (pressure, S/F, effort)

Group	Reports
Ventilation-first	Caught and lead by share of pre-escalation hours with oxygen recorded (< 25%, 25-50%, 50-75%, > 75%); caught by the hospital's oxygen-charting band (rarely < 10% of hours: 54 hospitals, 283 patients; sometimes 10-30%: 35, 201; often 30-60%: 34, 86; mostly > 60%: 76, 276), for the frozen alarm, S/F-forward A and B, and NEWS; first reading to speak
Recovered without escalation	False ambers and reds per recovered patient-day; ambers per 20-bed, 12-hour shift (frozen alarm and NEWS)
Early crash	Caught before escalation and lead, overall and by route (vasopressor first, both, ventilation first)
Combined	Caught and lead, frozen alarm vs NEWS (about 42 stays: too few to interpret)
All deteriorating groups	Share with no amber before escalation, by group

Run procedure and deviations

The eICU run happens once, after this document is signed, tagged and posted, using code already committed and tested.

1. Sign: commit, tag `reg2a-signed`, post this document and the code to OSF.
2. `eicu_cast_tip.sql` -> `eicu_cast.csv` (validated by dry run before signing).
3. `eicu_hourly.sql` -> `eicu_hourly.csv` (validated by dry run before signing).
4. `python3 eicu_run.py` -> `eicu_out.txt` (tested end to end on synthetic data; 0 errors).
5. Commit the output unedited and tag `reg2a-results`.
6. Read the primary verdict first, then each secondary and report in the order written here.

Deviations: any change forced by the run (a query failure, a data format surprise) is written down with its date, reason and effect before any result is read, and reported with the results. Nothing is fixed silently, and no analysis added afterwards is called pre-specified.

Search log

Every version tried on MIMIC-III before freezing is listed here, including the failures, because choosing the best of many on the same data flatters the winner.

Check	Version tried	Outcome
C14b-C14f	Averaged pressure road vs NEWS-lite	Kept idea; replaced by leading channel
C14d	Road OR NEWS combined	Set aside (no gain)
C16d	Leading channel 3-h vs averaged road	Adopted
C16f	Own 3-h line per pressure channel	Set aside (no gain)
C16g	Effort witness splitting the S/F budget	Set aside
C16i	Breathing fallback (S/F or effort)	Kept as S/F-forward B secondary
C18	Red by dwell or one-hour jump	Set aside (worse than NEWS)
C18b	Red by agreement with NEWS	Set aside (no gain)
C24b	20 versions: pressure share 0.4-0.8, effort A/B, gate on/off	Chose B, gate off, pressure 0.7 (best total on thinned data)
C21	Re-check red (found after the search was declared stopped)	Kept as key secondary

Integrity checks that lowered results: corrected tip (1,502 -> 1,375 deteriorating warned); eICU-like oxygen thinning (1,375 -> 1,176-1,216).

Where to verify: each check's rule (written before it ran, dated by its commit) and recorded result are in `MIMIC3_COVERAGE_RULE.md`; its script and printed output are listed in `CALIBRATION_INDEX.md`, with outputs in `calibration/`, in the project repository (github.com/Aditya0606-max/missh-public).

Limits and pre-mortem

The expected strength is narrow: more vasopressor-first patients warned than NEWS at the same cost, at about the same time, not earlier.

Known limits

- Not earlier than NEWS on the clock; the gain is more patients warned.

- Ventilation-first group: NEWS is expected ahead for the frozen alarm (it reads consciousness and oxygen use; eICU lacks oxygen information in 56% of hours, and the frozen S/F line is strict).
- Early-crash group: the fall happens inside the hour; hourly values see little (about 40 minutes of warning in rehearsal).
- Late crossers and the combined group are caught late or not at all; many combined patients are uncharted before the tube.
- eICU oxygen charting is lopsided (oxygen information in a median 19% of recovered patients′ hours vs 86% of vasopressor-first patients′ hours) and partly reflects clinician concern; the effort-only check is pre-set to show its effect.
- eICU′s 5-minute vitals are averaged hourly; within-hour change is unused.
- Vasopressor and ventilator start times are clinician decisions, not body events. Sepsis is defined by ICU admission diagnosis, not by Sepsis-3 criteria.

Pre-mortem: if the primary fails, the likely reasons

- eICU hospitals start vasopressors at different pressures than MIMIC-III′s hospital, shifting where the shock end sits.
- Pulse pressure and shock index behave differently with pooled cuff and arterial readings.
- The lines, chosen as the best of 20 versions on MIMIC-III, were flattered by that selection.
- Recovered patients′ pseudo-escalation times, drawn from eICU′s deteriorating patients, give a different watch length than in MIMIC-III.

MIMIC-III rehearsal results (not evidence)

On MIMIC-III as charted, the frozen alarm warned 79.8% of vasopressor-first patients vs full NEWS′s 64.3% at matched cost (105 vs 32 discordant); these numbers come from the data the alarm was tuned on and only show the pipeline works.

Report	Frozen alarm	Full NEWS (matched cost)
False ambers per recovered patient-day	0.87	0.86
Vasopressor-first warned (median lead)	79.8% (8.1 h)	64.3% (9.1 h)
Ventilation-first warned	49%	58%

Report	Frozen alarm	Full NEWS (matched cost)
Combined warned	34%	29%
Whole-ward total	1,182	1,203
Vasopressor-first, shock index stayed low	59%	42%
Late crossers	48% (2.0 h)	37%
Re-check red, vasopressor-first (red cost 0.30 per recovered patient-day)	47%	35% (NEWS alone); ~35-36% (NEWS re-check, interpolated)
Ambers per 20-bed, 12-h shift	9.6	9.5
Alarms starting while improving	44%	46%
Early crash warned (pressure and effort only)	32% (0.7 h)	-

S/F-forward A and B (costs 0.99 and 0.95) warned 65% and 64% of the ventilation-first group vs full NEWS's 60% at 0.92, and 76% and 75% of the vasopressor-first group vs 67%. On MIMIC-III thinned to eICU's oxygen charting, the whole-ward total was 1,216 vs NEWS-lite 1,246, with the vasopressor-first group at 80% vs 73%. In paired timing, the two alarms usually spoke in the same hour (median difference 0 h); S/F-forward B spoke first for 38% of ventilation-first patients warned by both, NEWS for 31%. Precision was 42% for the frozen alarm, 40% for B and 43% for full NEWS.

Sign-off

Signing confirms that no eICU outcome has been viewed, that every number above was fixed before the run, and that the result will be published whatever it shows.

- [ ] Read end to end; every section agrees with the committed code
- [ ] Commit tagged `reg2a-signed`
- [ ] Document and code posted to OSF (public timestamp)
- [ ] eICU run once; output committed unedited and tagged `reg2a-results`

Signed by: Aditya, independent researcher. Date of signing: _____